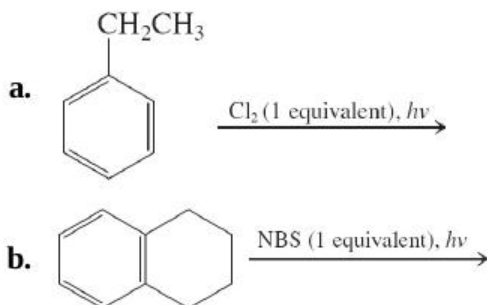
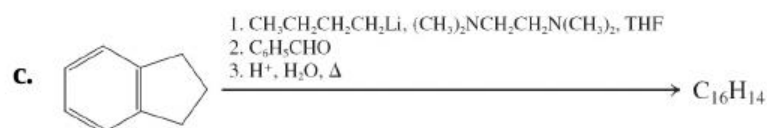
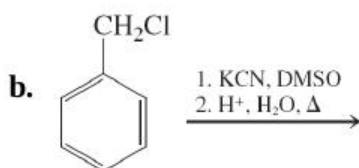
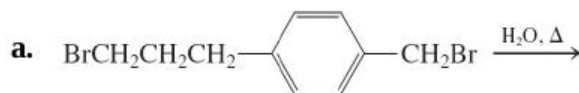


Problems

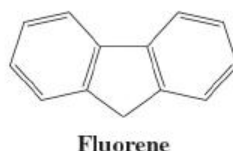
37. Give the expected major product(s) of each of the following reactions.



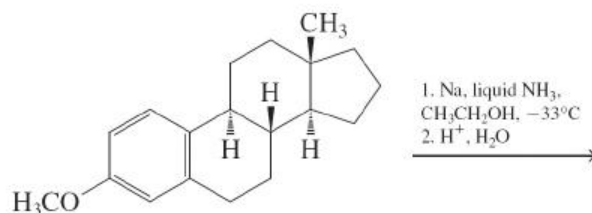
38. Formulate a mechanism for the reaction described in [Problem 37](#), part (b).
39. Propose syntheses of each of the following compounds, beginning in each case with ethylbenzene. **(a)** (1-Chloroethyl)- benzene; **(b)** 2-phenylpropanoic acid; **(c)** 2-phenylethanol; **(d)** 2-phenyloxacyclopropane.
40. Predict the order of relative stability of the three benzylic cations derived from chloromethylbenzene (benzyl chloride), 1-(chloromethyl)-4-methoxybenzene (4-methoxybenzyl chloride), and 1-(chloromethyl)-4-nitrobenzene (4-nitrobenzyl chloride). Rationalize your answer with the help of resonance structures.
41. By drawing appropriate resonance structures, illustrate why halogen atom attachment at the para position of the phenylmethyl (benzyl) radical is disfavored compared with attachment at the benzylic position.
42. Triphenylmethyl radical, $(\text{C}_6\text{H}_5)_3\text{C}\cdot$, is stable at room temperature in dilute solution in an inert solvent, and salts of triphenylmethyl cation, $(\text{C}_6\text{H}_5)_3\text{C}^+$, can be isolated as stable crystalline solids. Propose explanations for the unusual stabilities of these species.
43. Give the expected products of the following reactions or reaction sequences.



44. The hydrocarbon with the common name fluorene is acidic enough ($\text{p}K_a \approx 23$) to be a useful indicator in deprotonation reactions of compounds of greater acidity. Indicate the most acidic hydrogen(s) in fluorene. Draw resonance structures to explain the relative stability of its conjugate base.



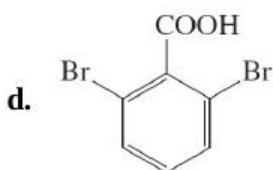
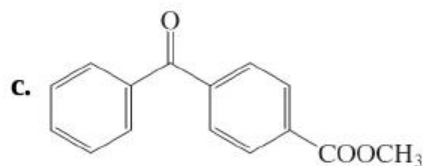
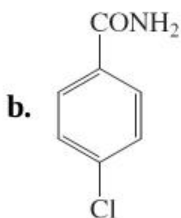
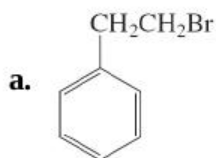
45. The Birch reduction shown below is widely used in the synthesis of a variety of highly biologically active steroids. Although the ^1H NMR spectrum of its product is complex, it exhibits a singlet at $\delta = 5.73$ ppm integrating to one hydrogen. In addition, the IR spectrum shows absorptions at $\tilde{\nu} = 1620$ and 1660 cm^{-1} , of medium and very strong intensities, respectively.



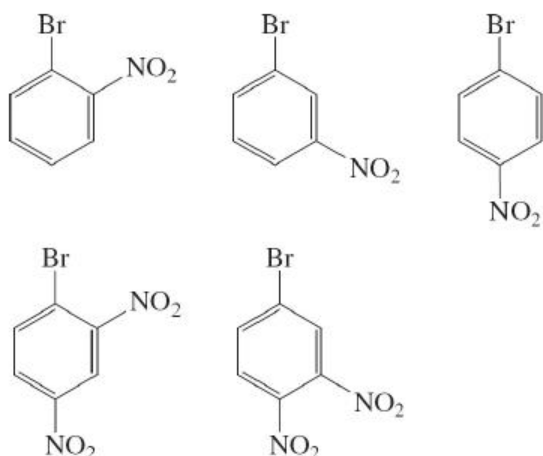
Propose a structure for the product and rationalize its formation

mechanistically.

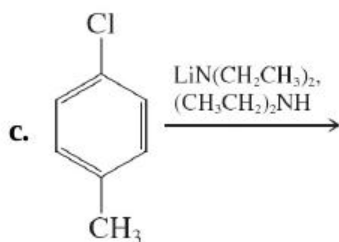
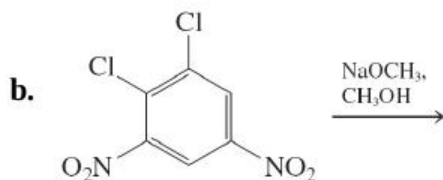
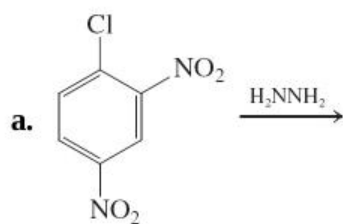
46. Outline a straightforward, practical, and efficient synthesis of each of the following compounds. Start with benzene or methylbenzene. Assume that the para isomer (but not the ortho isomer) may be separated efficiently from any mixtures of ortho and para substitution products. Work backward from the target.



47. Rank the following molecules in descending order of reactivity toward hydroxide ion.

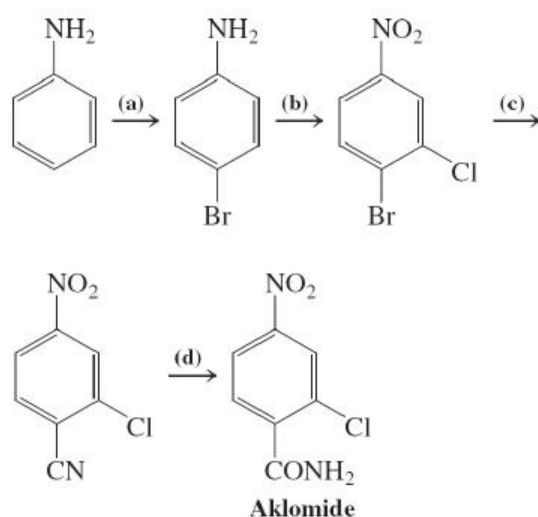


48. Predict the main product(s) of the following reactions. In each case, describe the mechanism(s) in operation.

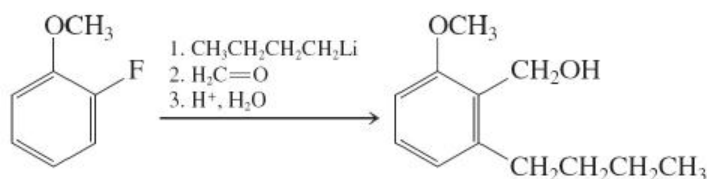


49. Starting with aniline, propose a synthesis of aklomide, an agent used to treat certain exotic fungal and protozoal infections in veterinary medicine. Several intermediates are shown to give you the general route.

Fill in the blanks that remain; each requires as many as three sequential reactions. (**Hint:** Review the oxidation of amino- to nitroarenes in [Section 16-5.](#))

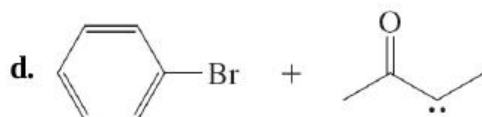
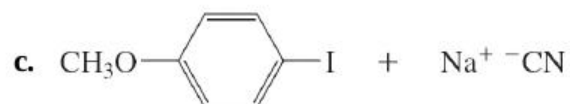
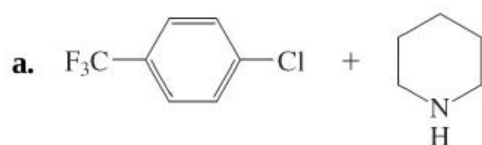


50. Explain the mechanism of the following synthetic transformation. (**Hint:** Two equivalents of butyllithium are used.)

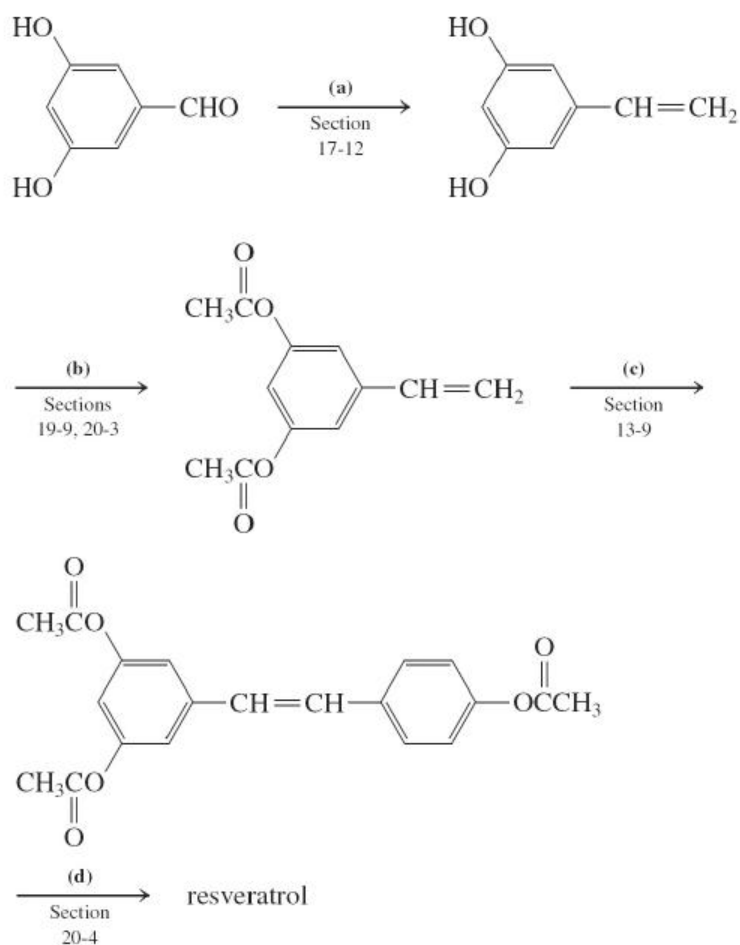


51. In nucleophilic aromatic substitution reactions that proceed by the addition–elimination mechanism, fluorine is the most easily replaced halogen in spite of the fact that $\text{F}-\text{F}^-$ is by far the worst leaving group among halide ions. For example, 1-fluoro-2,4-dinitrobenzene reacts much more rapidly with amines than does the corresponding chloro compound. Suggest an explanation. (**Hint:** Consider the effect of the identity of the halogen on the rate-determining step.)
52. Based on the mechanism presented for the Pd-catalyzed reaction of a halobenzene with hydroxide ion, write out a reasonable mechanism for the Pd-catalyzed reaction of 1-bromo-3-methoxybenzene with 2-methyl-1-propanamine to generate 3-methoxy-*N*-(2-methylpropyl)aniline, shown in [Section 22-4.](#)

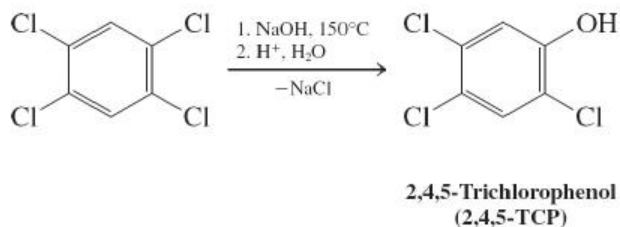
53. Give the likely products of each of the following reactions. Each one is carried out in the presence of a Pd catalyst, a phosphine, and heat.

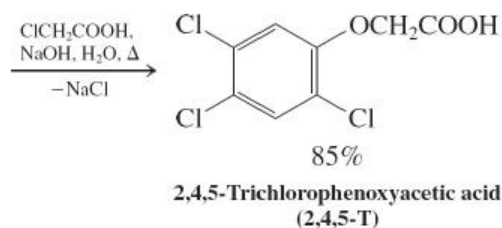


54. A very efficient short synthesis of resveratrol ([Real Life 22-1](#)) was reported in 2006. Fill in reasonable reagents for steps (a) through (d). Refer to the referenced text sections as needed.

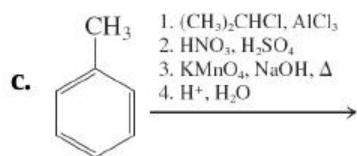
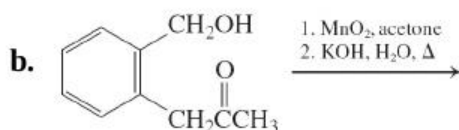
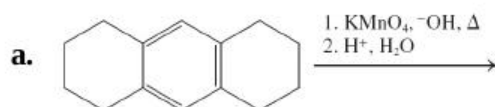


55. The reaction sequence shown below and on the next page illustrates the synthesis of 2,4,5-trichlorophenoxyacetic acid (2,4,5-T), a powerful herbicide. A 1 : 1 mixture of the butyl esters of 2,4,5-T and its dichlorinated analog 2,4-D was used between 1965 and 1970 as a defoliant during the Vietnam War under the code name Agent Orange. Propose mechanisms for the reactions in the synthesis of this substance, whose effects on the health of those exposed to it remain topics of considerable controversy.

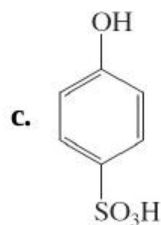


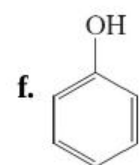
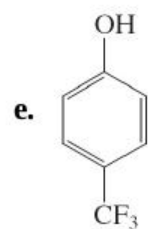
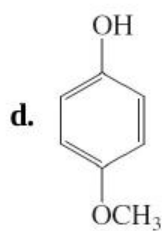


56. Give the expected major product(s) of each of the following reactions and reaction sequences.

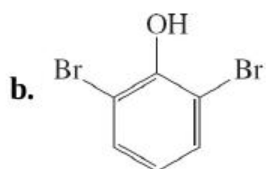
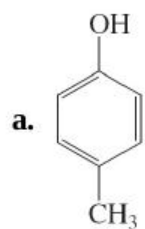


57. Rank the following compounds in order of descending acidity.

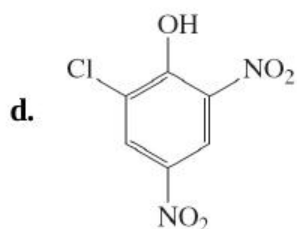




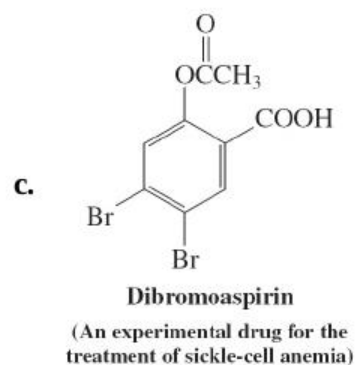
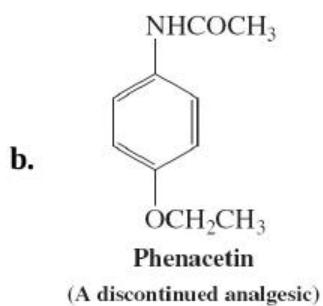
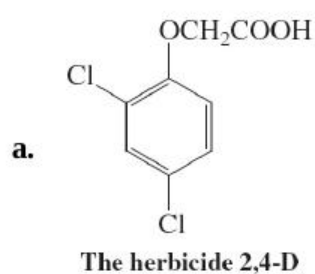
58. Design a synthesis of each of the following phenols, starting with either benzene or any monosubstituted benzene derivative.

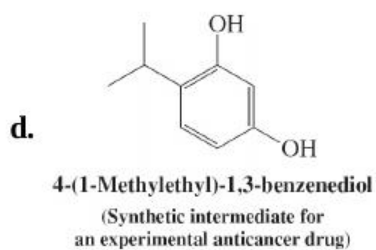


c. The three benzenediols

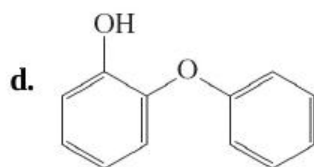
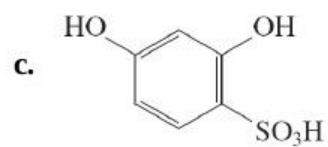
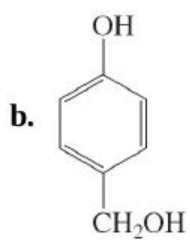
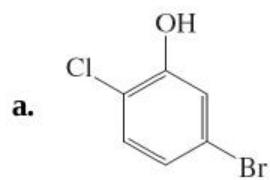


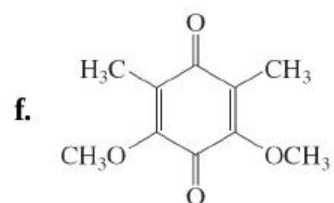
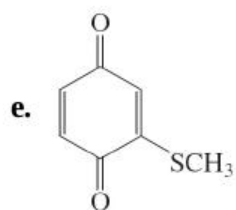
59. Starting with benzene, propose syntheses of each of the following phenol derivatives.



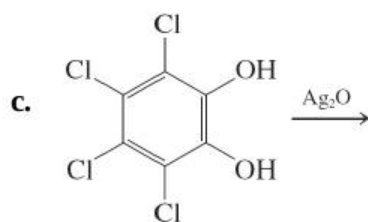
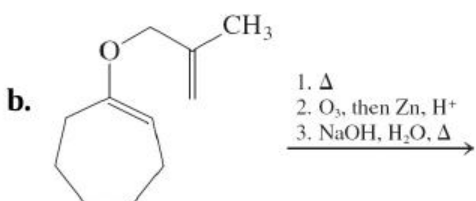
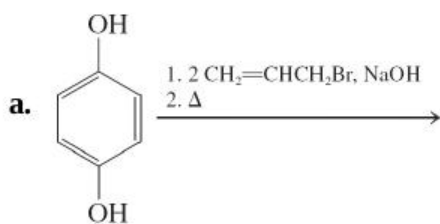


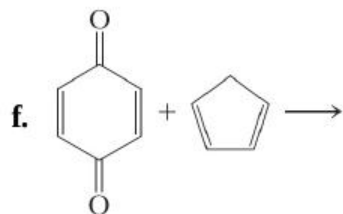
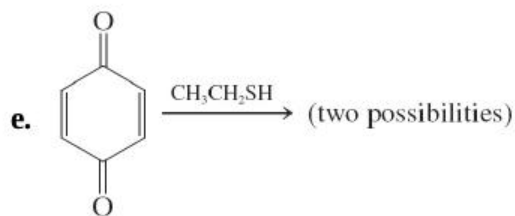
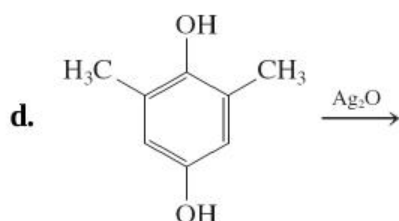
60. Name each of the following structures.





61. Give the expected product(s) of each of the following reaction sequences.





62. As a children's medicine, acetaminophen (Tylenol) has a major marketing advantage over aspirin: Liquid Tylenol preparations (essentially, acetaminophen dissolved in flavored water) are stable, whereas comparable aspirin solutions are not. Indeed, phenyl alkanoates undergo hydrolysis (and transesterification) considerably more rapidly than alkyl alkanoates, reactivity that underlies the mechanism of action of aspirin ([Real Life 22-2](#)). Explain.
63. Chili peppers contain significant quantities of vitamins A, C, and E, as well as folic acid and potassium. They also contain small quantities of capsaicin, the fiery essence responsible for the "hot" in hot peppers (see [p. 1081](#)). The pure substance is in fact quite dangerous: Chemists handling capsaicin must work in special filtered-air enclosures and wear full body protection. One milligram placed on the skin will cause a severe burn. Although capsaicin has no odor or flavor of its own, its pungency—in the form of stimulation of the nerves in the mucous membranes of the mouth—can be detected even when the substance is diluted to one part in 17 million of water. The hottest peppers exhibit about 1/20th of this level of pungency.

The structure of capsaicin is shown on [p. 1080](#). Some of the data that were used in its elucidation are presented below. Interpret as much of this information as you can.

MS: $m/z = 305$ (M^+), MS : $m/z = 305$ (M^+), 195 (tricky!), 137 (base peak), 122.

IR: $\tilde{\nu} = 972, 1660, 3016, 3445, 3541$ cm^{-1} .

IR : $\tilde{\nu} = 972, 1660, 3016, 3445, 3541$ cm^{-1} .

$^1\text{H NMR}$: $\delta = 0.93$ (d, $J = 8$ Hz, 6 H),

$^1\text{H NMR}$: $\delta = 0.93$ (d, $J = 8$ Hz, 6 H), 1.35 (quin, $J = 7$ Hz, 2 H),

1.35 (quin, $J = 7$ Hz, 2 H), 1.62 (quin, $J = 7$ Hz, 2 H),

1.62 (quin, $J = 7$ Hz, 2 H), 1.97 (q, $J = 7$ Hz, 2 H), 1.97 (q, $J = 7$ Hz, 2 H),

2.18 (t, $J = 7$ Hz, 2 H), 2.18 (t, $J = 7$ Hz, 2 H), 2.20 (m, 1 H), 3.85 (s, 3H),

2.20 (m, 1 H), 3.85 (s, 3H), 4.33 (d, $J = 6$ Hz, 2 H), 4.33 (d, $J = 6$ Hz, 2 H),

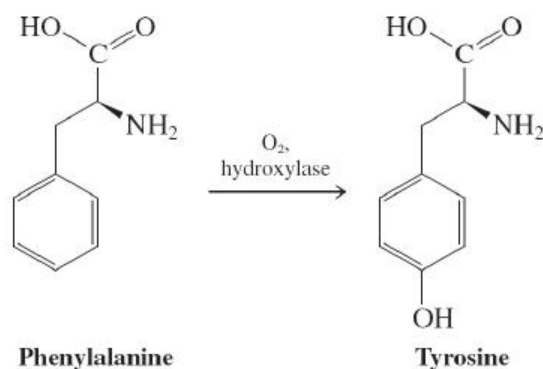
5.33 (m, 2 H), 5.82 (broad s, 2 H), 5.33 (m, 2 H), 5.82 (broad s, 2 H), 6.73

(dd, $J = 8, 2$ Hz, 1 H), 6.73 (dd, $J = 8, 2$ Hz, 1 H), 6.78 (d, $J = 2$ Hz, 1 H),

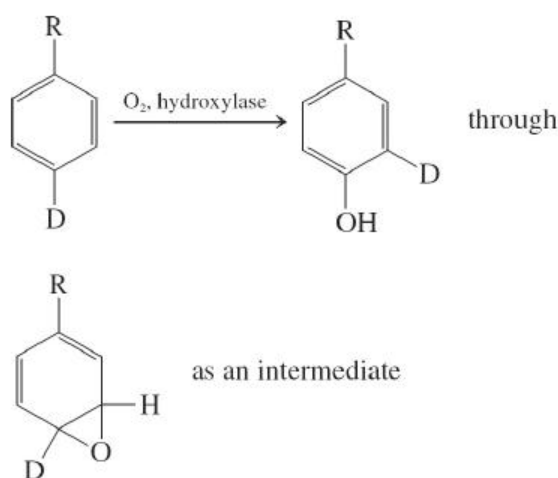
6.78 (d, $J = 2$ Hz, 1 H), 6.85 (d, $J = 8$ Hz, 1 H) ppm.

6.85 (d, $J = 8$ Hz, 1 H) ppm.

64. **CHALLENGE** Biochemical oxidation of aromatic rings is catalyzed by a group of liver enzymes called aryl hydroxylases. Part of this chemical process is the conversion of toxic aromatic hydrocarbons such as benzene into water-soluble phenols, which can be easily excreted. However, the primary purpose of the enzyme is to enable the synthesis of biologically useful substances, like the amino acid tyrosine from its relative phenylalanine (below).



(a) Extrapolating from your knowledge of benzene chemistry, which of the following three possibilities seems most reasonable: The oxygen is introduced by electrophilic attack on the ring; the oxygen is introduced by free-radical attack on the ring; or the oxygen is introduced by nucleophilic attack on the ring? **(b)** It is widely suspected that oxacyclopropanes play a role in arene hydroxylation. Part of the evidence is the following observation: When the site to be hydroxylated is labeled with deuterium, a substantial proportion of the product still contains deuterium atoms, which have apparently migrated to the position ortho to the site of hydroxylation.

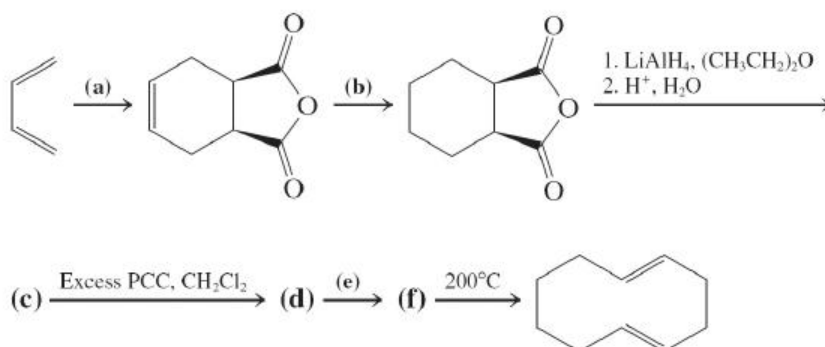


Suggest a plausible mechanism for the formation of the oxacyclopropane intermediate and its conversion into the observed product. (**Hint:** Hydroxylase converts O_2 into hydrogen peroxide, $\text{HO}-\text{OH}$.) Assume the availability of catalytic amounts of acids and bases, as necessary.

Note: In victims of the genetically transmitted disorder called phenylketonuria (PKU), the hydroxylase enzyme system described here does not function properly. Instead, phenylalanine in the brain is converted into 2-phenyl-2-oxopropanoic (phenylpyruvic) acid, the reverse of the process shown in [Problem 57](#) of [Chapter 21](#). The buildup of this species in the brain can lead to severe retardation; therefore, people with PKU (which can be diagnosed at birth) must be restricted to diets low in phenylalanine.

65. A common application of the Cope rearrangement is in ringenlargement

sequences. Fill in the reagents and products missing from the following scheme, which illustrates the construction of a 10-membered ring.



66. As mentioned in [Section 22-10](#), the Sandmeyer reactions, which entail the substitution of the nitrogen in arenediazonium salts by Cl, Br, CN , or I , require copper(I) ion as a catalyst and proceed by complex radical mechanisms. Why do such substitutions not follow an $\text{S}_{\text{N}}2$ pathway? Why is the $\text{S}_{\text{N}}1$ mechanism, which operates in the corresponding displacements with OH^- and I^- , not effective?
67. Formulate a detailed mechanism for the diazotization of aniline in the presence of HCl and NaNO_2 . Then suggest a plausible mechanism for its subsequent conversion into iodobenzene by treatment with aqueous iodide ion (e.g., from K^+I^-). Bear in mind your answer to [Problem 66](#).
68. Show how you would convert 3-methylaniline into each of the following compounds: **(a)** methylbenzene; **(b)** 1-bromo-3-methylbenzene; **(c)** 3-methylphenol; **(d)** 3-methylbenzonitrile; **(e)** *N*-ethyl-3-methylaniline.
69. Devise a synthesis of each of the following substituted benzene derivatives, starting from benzene.

